

Day 12 Hybrid Notes

45-45-90 and 30-60-90 triangle patterns | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets
I can identify 45-45-90 and 30-60-90 triangles.
I can use side ratios to find missing lengths.
I can combine special right triangle patterns with coordinate geometry.

Part 1: Cloze Notes - Essential Ideas
1. A 45-45-90 triangle has two congruent _____.
2. The side ratio in a 45-45-90 triangle is $x : x :$ _____.
3. A 30-60-90 triangle has angle measures 30, 60, and _____.
4. The side ratio in a 30-60-90 triangle is $x : x\sqrt{3} :$ _____.
5. The shortest side is opposite the _____.
6. The hypotenuse is opposite the _____.
7. Exact answers often include _____.

Part 2: Outline Notes - Big Picture A. Special right triangles are shortcuts from the Pythagorean _____. B. Angle measures determine side _____. C. Ratios help prepare for _____. D. Sketching and labeling prevent side _____. _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ —
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Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

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Guided setup + active learning

Part 3: Guided Setup Practice

1. In a 45-45-90 triangle, one leg is 7. Find the hypotenuse.

Setup/work: _____

Answer: _____

2. In a 30-60-90 triangle, the short leg is 5. Find the hypotenuse and long leg.

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student puts $\sqrt{3}$ across from 30 degrees.
What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A square has side length 9. Find its diagonal exactly.

Setup: _____

Answer: _____

Sentence answer: _____